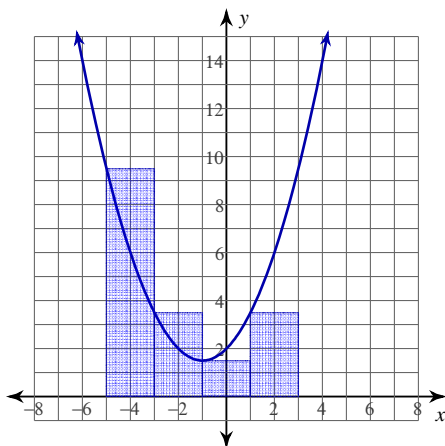


Approximating Area Under a Curve

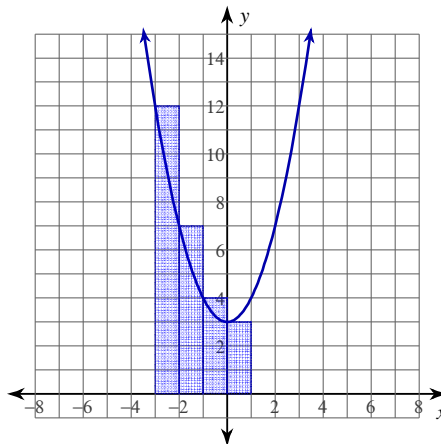
Date _____ Period _____

For each problem, approximate the area under the curve over the given interval using 4 left endpoint rectangles.

1) $y = \frac{x^2}{2} + x + 2$; $[-5, 3]$

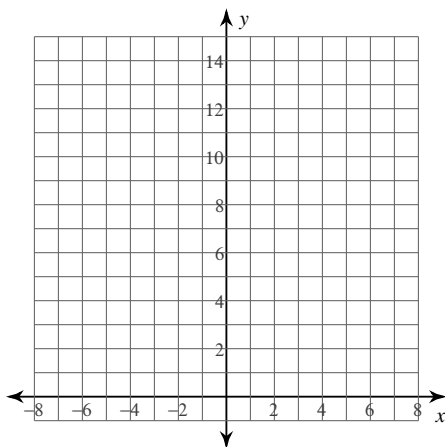


2) $y = x^2 + 3$; $[-3, 1]$



For each problem, approximate the area under the curve over the given interval using 5 right endpoint rectangles. You may use the provided graph to sketch the curve and rectangles.

3) $y = -\frac{x^2}{2} + 6$; $[-3, 2]$



4) $y = -\frac{x^2}{2} + x + 5$; $[-1, 4]$

